

Template Week 2 – Logic

Student number: 587890

Assignment 2.1: Parking lot

Which gates do you need?

Nand after that a not

Complete this table

Parking lot 1	Parking lot 2	Parking lot 3	Result (full)
0	0	0	0
0	0	1	0
0	1	0	0
1	0	0	0
0	1	1	0
1	0	1	0
1	1	0	0
1	1	1	0

Assignment 2.2: Android or iPhone

Which gates do you need?

NOR

Complete this table

Android phone	iPhone	Result (Phone in possession)
0	0	0
1	0	1
0	1	1
1	1	1

Assignment 2.3: Four NAND gates

Complete this table

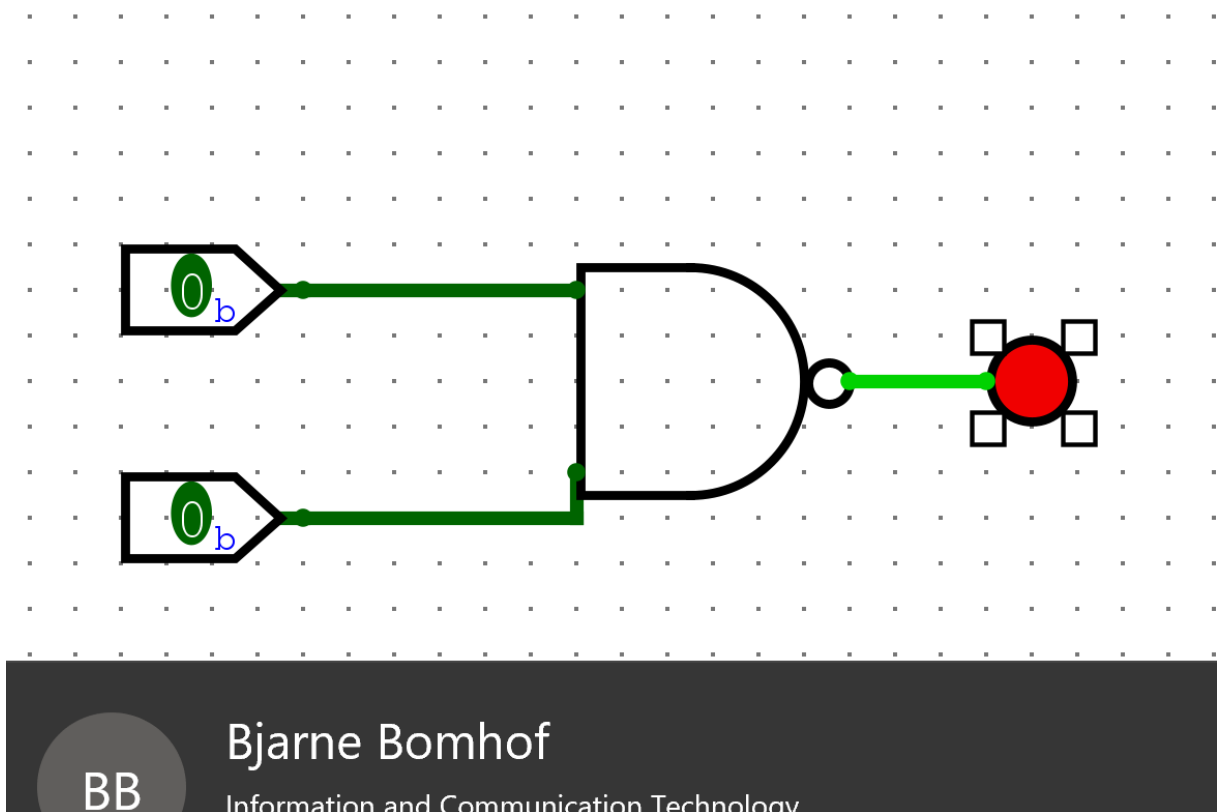
A	B	Q
0	0	0
0	1	1
1	0	1
1	1	0

How can the design be simplified?

Just use a xor gate and a not gate

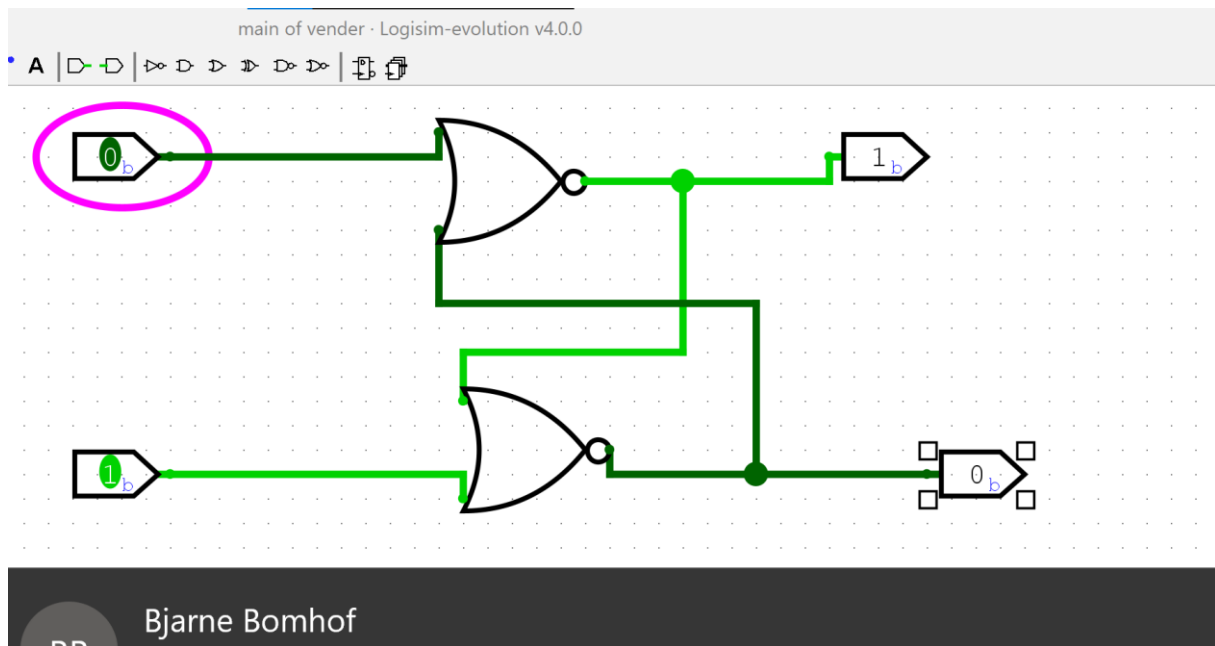
Assignment 2.4: Getting to know Logisim evolution

Screenshot of the design with your name and student number in it:



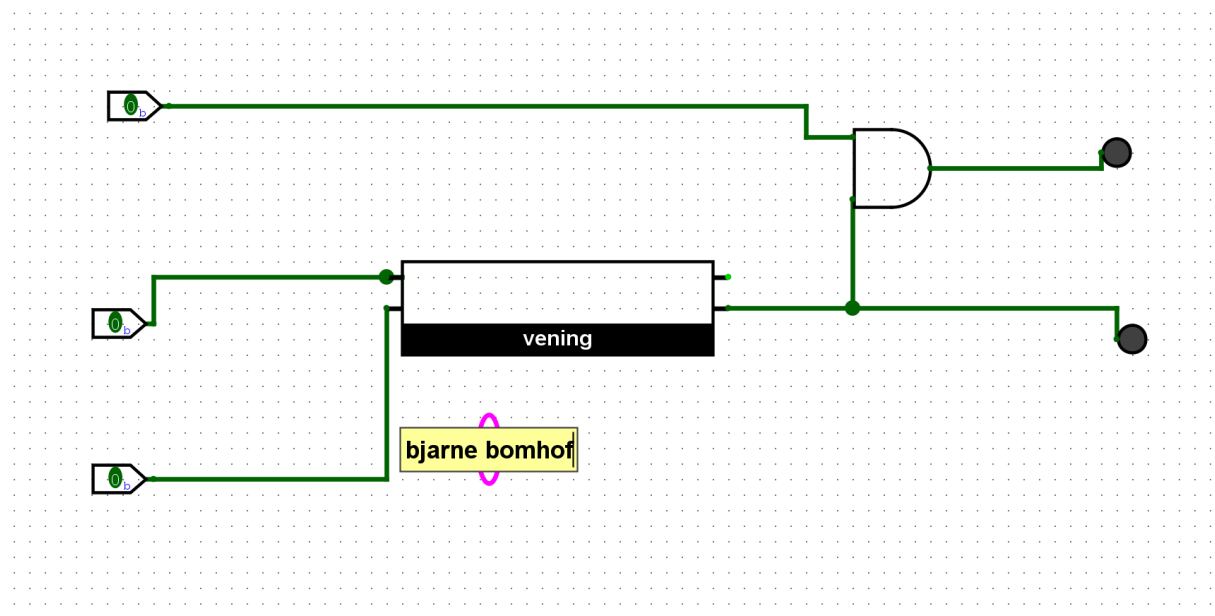
Assignment 2.5: SR Latch

Screenshot SR Latch in Logisim with your name and student number:



Assignment 2.6: Vending Machine

Screenshot Vending Machine in Logisim with your name and student number:



Assignment 2.7: Bitwise operators

Complete the java source code for bitwise operators. Put the source code here.

```
public class Main {  
    public static void main(String[] args) {  
        int number = 5;  
  
        // Check odd/even using bitwise AND  
        if ((number & 1) == 1)  
            System.out.println("number is odd");  
        else  
            System.out.println("number is even");  
    }  
}
```

Assignment 2.8: Java Application Bit Calculations

Create a java program that accepts user input and presents a menu with options.

1. Is number odd?
2. Is number a power of 2?
3. Two's complement of number?

Implement the methods by using the bitwise operators you have just learned.

Organize your source code in a readable manner with the use of control flow and methods.

Keep this application because you need to expand it in week 6 for calculating network segments.

Paste source code here, with a screenshot of a working application.

```
import java.util.Scanner;  
  
public class BitwiseMenuProgram {  
    public static boolean isOdd(int number) {  
        return (number & 1) == 1;  
    }  
}
```

```

public static boolean isPowerOfTwo(int number) {
    return number > 0 && (number & (number - 1)) == 0;
}

public static int twosComplement(int number) {
    return (~number) + 1;
}

public static void displayMenu() {
    System.out.println("\n--- Bitwise Operations Menu ---");
    System.out.println("1. Is number odd?");
    System.out.println("2. Is number a power of 2?");
    System.out.println("3. Two's complement of number");
    System.out.println("4. Exit");
    System.out.print("Enter your choice: ");
}

public static void main(String[] args) {
    Scanner scanner = new Scanner(System.in);
    int choice;

    do {
        displayMenu();
        choice = scanner.nextInt();

        if (choice >= 1 && choice <= 3) {
            System.out.print("Enter an integer: ");
            int number = scanner.nextInt();

            switch (choice) {

```

```

case 1:

    System.out.println(number + " is " + (isOdd(number) ? "odd" : "even"));

    break;

case 2:

    System.out.println(number + " is " +

        (isPowerOfTwo(number) ? "a power of 2" : "not a power of 2"));

    break;

case 3:

    System.out.println("Two's complement of " + number + " is " +

        twosComplement(number));

    break;

}

} else if (choice != 4) {

    System.out.println("Invalid choice. Please try again.");

}

} while (choice != 4);

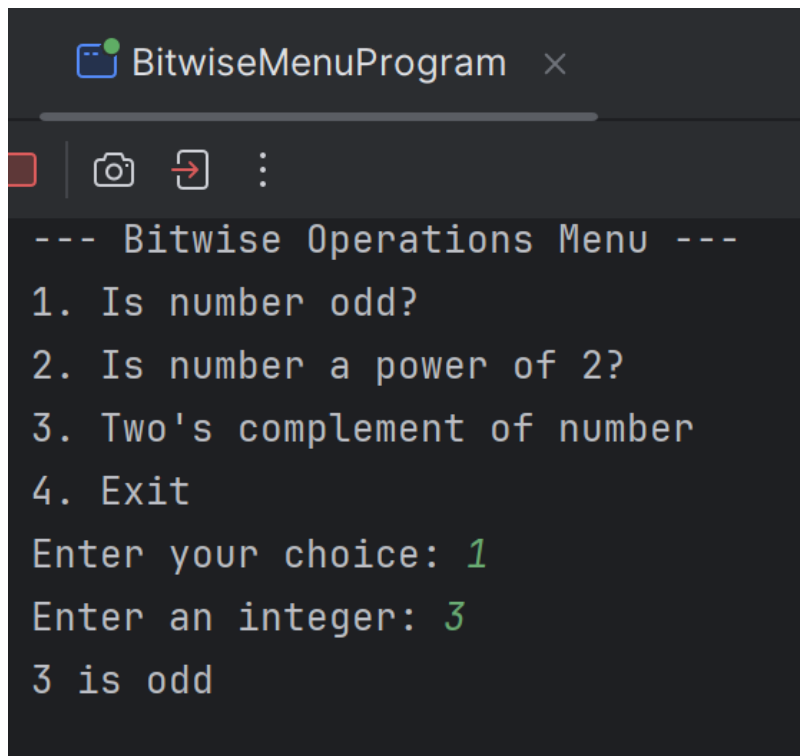
System.out.println("Program terminated.");

scanner.close();

}

}

```



The screenshot shows a window titled "BitwiseMenuProgram" with a close button. Below the title bar is a toolbar with a red square, a camera icon, a right-pointing arrow icon, and a vertical ellipsis icon. The main content area displays a menu titled "--- Bitwise Operations Menu ---" with four options: "1. Is number odd?", "2. Is number a power of 2?", "3. Two's complement of number", and "4. Exit". Below the menu, the user is prompted to "Enter your choice:" and the number "1" is entered. Then, the user is prompted to "Enter an integer:" and the number "3" is entered. Finally, the output "3 is odd" is displayed.

```
--- Bitwise Operations Menu ---
1. Is number odd?
2. Is number a power of 2?
3. Two's complement of number
4. Exit
Enter your choice: 1
Enter an integer: 3
3 is odd
```

Ready? Then save this file and export it as a pdf file with the name: [week2.pdf](#)